JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

IIT JAM 2009 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
15	90 M	60 Min	MCQ

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	15	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2009 • Q1 • ID: JAM_2009_Q1) MCQ +6 M Neg: −2.0

Q.1 Let V be the vector space of all 6×6 real matrices over the field $\mathbb{R}$. Then the dimension of the subspace of V consisting of all symmetric matrices is

(A) 15 (B) 18 (C) 21 (D) 35

Question 2 (JAM 2009 • Q2 • ID: JAM_2009_Q2) MCQ +6 M Neg: −2.0

Q.2 Let R be the ring of all functions from $\mathbb{R}$ to $\mathbb{R}$ under point-wise addition and multiplication. Let $I = \{f : \mathbb{R} \to \mathbb{R} \mid f \text{ is a bounded function}\}$, $J = \{f : \mathbb{R} \to \mathbb{R} \mid f(3) = 0\}$. Then

(A) J is an ideal of R but I is not an ideal of R
(B) I is an ideal of R but J is not an ideal of R
(C) both I and J are ideals of R
(D) neither I nor J is an ideal of R

Question 3 (JAM 2009 • Q3 • ID: JAM_2009_Q3) MCQ +6 M Neg: −2.0

Q.3 Which of the following sequences of functions is uniformly convergent on $(0,1)$?

(A) x^n (B) $\dfrac{n}{nx+1}$ (C) $\dfrac{x}{nx+1}$ (D) $\dfrac{1}{nx+1}$

Question 4 (JAM 2009 • Q4 • ID: JAM_2009_Q4) MCQ +6 M Neg: −2.0

Q.4 Let $T : \mathbb{R}^4 \to \mathbb{R}^4$ be a linear transformation satisfying $T^3 + 3T^2 = 4I$, where I is the identity transformation. Then the linear transformation $S = T^4 + 3T^3 - 4I$ is

(A) one-one but not onto
(B) onto but not one-one
(C) invertible
(D) non-invertible

Question 5 (JAM 2009 • Q5 • ID: JAM_2009_Q5) MCQ +6 M Neg: −2.0

Q.5 The number of all subgroups of the group $(Z_{60}, +)$ of integers modulo 60 is

(A) 2 (B) 10 (C) 12 (D) 60

Question 6 (JAM 2009 • Q6 • ID: JAM_2009_Q6) MCQ +6 M Neg: −2.0

Q.6 Let $a_n = \begin{cases} \dfrac{1}{3^n} & \text{if } n \text{ is a prime,} \\ \dfrac{1}{4^n} & \text{if } n \text{ is not a prime.} \end{cases}$

Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ is

(A) 4 (B) 3 (C) $\frac{1}{3}$ (D) $\frac{1}{4}$

Question 7 (JAM 2009 • Q7 • ID: JAM_2009_Q7) MCQ +6 M Neg: −2.0

Q.7 The set of all limit points of the sequence $1, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}, \frac{1}{16}, \frac{3}{16}, \frac{5}{16}, \frac{7}{16}, \frac{9}{16}, \ldots$ is

(A) $[0, 1]$

(B) $(0, 1]$

(C) the set of all rational numbers in $[0, 1]$

(D) the set of all rational numbers in $[0, 1]$ of the form $\frac{m}{2^n}$ where m and n are integers

Question 8 (JAM 2009 • Q8 • ID: JAM_2009_Q8) MCQ +6 M Neg: −2.0

Q.8 Let $F: \mathbb{R} \to \mathbb{R}$ be a continuous function and $a > 0$. Then the integral $\int_0^a \left[\int_0^x F(y)\, dy \right] dx$ equals

(A) $\int_0^a y F(y)\, dy$

(B) $\int_0^a (a-y) F(y)\, dy$

(C) $\int_0^a (y-a) F(y)\, dy$

(D) $\int_a^0 y F(y)\, dy$

Question 9 (JAM 2009 • Q9 • ID: JAM_2009_Q9) MCQ +6 M Neg: −2.0

Q.9 The set of all positive values of a for which the series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \tan^{-1}\left(\frac{1}{n}\right) \right)^a$ converges, is

(A) $\left(0, \frac{1}{3}\right]$

(B) $\left(0, \frac{1}{3}\right)$

(C) $\left[\frac{1}{3}, \infty\right)$

(D) $\left(\frac{1}{3}, \infty\right)$

Question 10 (JAM 2009 • Q10 • ID: JAM_2009_Q10) MCQ +6 M Neg: −2.0

Q.10 Let a be a non-zero real number. Then $\lim_{x \to a} \frac{1}{x^2 - a^2} \int_a^x \sin(t^2)\, dt$ equals

(A) $\frac{1}{2a} \sin(a^2)$

(B) $\frac{1}{2a} \cos(a^2)$

(C) $-\frac{1}{2a} \sin(a^2)$

(D) $-\frac{1}{2a} \cos(a^2)$

Question 11 (JAM 2009 • Q11 • ID: JAM_2009_Q11) MCQ +6 M Neg: −2.0

Q.11 Let $T(x, y, z) = xy^2 + 2z - x^2 z^2$ be the temperature at the point (x, y, z). The unit vector in the direction in which the temperature decreases most rapidly at $(1, 0, -1)$ is

(A) $-\frac{1}{\sqrt{5}} \hat{i} + \frac{2}{\sqrt{5}} \hat{k}$

(B) $\frac{1}{\sqrt{5}} \hat{i} - \frac{2}{\sqrt{5}} \hat{k}$

(C) $\frac{2}{\sqrt{14}} \hat{i} + \frac{3}{\sqrt{14}} \hat{j} + \frac{1}{\sqrt{14}} \hat{k}$

(D) $-\left(\frac{2}{\sqrt{14}} \hat{i} + \frac{3}{\sqrt{14}} \hat{j} + \frac{1}{\sqrt{14}} \hat{k} \right)$

Question 12 (JAM 2009 • Q12 • ID: JAM_2009_Q12) MCQ +6 M Neg: −2.0

Q.12 Consider the differential equation $2\cos(y^2)\,dx - xy\sin(y^2)\,dy = 0$. Then

(A) e^x is an integrating factor
(B) e^{-x} is an integrating factor
(C) $3x$ is an integrating factor
(D) x^3 is an integrating factor

Question 13 (JAM 2009 • Q13 • ID: JAM_2009_Q13) MCQ +6 M Neg: −2.0

Q.13 Suppose $\vec{V} = p(x, y)\hat{i} + q(x, y)\hat{j}$ is a continuously differentiable vector field defined in a domain D in $\mathbb{R}^2$. Which one of the following statements is NOT equivalent to the remaining ones?

(A) There exists a function $\phi(x, y)$ such that $\dfrac{\partial \phi}{\partial x} = p(x, y)$ and $\dfrac{\partial \phi}{\partial y} = q(x, y)$ for all $(x, y) \in D$

(B) $\dfrac{\partial q}{\partial x} = \dfrac{\partial p}{\partial y}$ holds at all points of D

(C) $\oint_C \vec{V} \cdot d\vec{r} = 0$ for every piecewise smooth closed curve C in D

(D) The differential $p\,dx + q\,dy$ is exact in D

Question 14 (JAM 2009 • Q14 • ID: JAM_2009_Q14) MCQ +6 M Neg: −2.0

Q.14 Let $f, g : [-1, 1] \to \mathbb{R}$, $f(x) = x^3$, $g(x) = x^2|x|$. Then

(A) f and g are linearly independent on $[-1, 1]$
(B) f and g are linearly dependent on $[-1, 1]$
(C) $f(x)g'(x) - f'(x)g(x)$ is NOT identically zero on $[-1, 1]$
(D) there exist continuous functions $p(x)$ and $q(x)$ such that f and g satisfy $y'' + py' + qy = 0$ on $[-1, 1]$

Question 15 (JAM 2009 • Q15 • ID: JAM_2009_Q15) MCQ +6 M Neg: −2.0

Q.15 The value of c for which there exists a twice differentiable vector field $\vec{F}$ with curl $\vec{F} = 2x\hat{i} - 7y\hat{j} + cz\hat{k}$ is

(A) 0 (B) 2 (C) 5 (D) 7

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2009 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+6	**C**
2	MCQ	+6	**A**
3	MCQ	+6	**C**
4	MCQ	+6	**D**
5	MCQ	+6	**C**
6	MCQ	+6	**B**
7	MCQ	+6	**A**
8	MCQ	+6	**B**

Q. No.	Type	Marks	Official Key
9	MCQ	+6	**C**
10	MCQ	+6	**A**
11	MCQ	+6	**B**
12	MCQ	+6	**D**
13	MCQ	+6	**B**
14	MCQ	+6	**A**
15	MCQ	+6	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.